VIDYAPEETH

BATCH CODE: 12-AN-181-MA-2026

SUBJECT NAME: PHYSICS

CHAPTER NAME:

UNITS & MEASUREMENT.

Lecture No. 12

By – Navneet Shrivastava

Today's Goal

Subtopic ..

ERROR

Subtopic ..

Subtopic ..

ERROR

INSTRUMENTAL

- Sources known

TYPES:-

1. ZERO ERROR
2. BACKLASH ERROR :- i) occurs in Screw guage. ii) Rachet should be rotated in only one direction.
3. LEAST COUNT ERROR (Permissible Error) :- ii) It is the min error made by an instrument which is equal to its LC (Least count)

RANDOM

- Sources unknown
- Can't be Eliminated Completely

1) meter scale = 1mm
VC = 0.1mm
SC = 0.01 mm

RANDOM ERROR :-

i) Error = $|MV - TV|$
= Always ⊕

(ii) TV does not exists
TV is taken as the avg. value of all the measurments

EXPRESSION OF ERROR :-

① ABSOLUTE ERROR :- Indivisual Error

$l_1 =$ $\quad \Delta a_1 = |l_1 - l_{av}|$ $\quad a_1 \rightarrow$ Abs. Error in measurment of l_1

$l_2 =$ $\quad \Delta a_2 = |l_2 - l_{av}|$ $\quad a_2 \rightarrow$ . . . . . l_2

$l_3 =$ $\quad \Delta a_3 = |l_3 - l_{av}|$

$\vdots$

$$l_{av} = \frac{l_1 + l_2 + \dots + l_n}{n} \qquad \Delta a_{mean} = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}$$

② MEAN ABSOLUTE Err

avg of all the absolute Errors

a_{mean}

③ Fractional Error / Relative Error. $= \dfrac{\Delta a_{mean}}{l_{mean}} = \dfrac{\text{Mean Abs Error}}{TV}$

④ % Error $= \dfrac{\Delta a_{mean}}{l_{mean}} \times 100$

Q:- 2.46m, 2.48m, 2.49m, 2.49m, and 2.46m

Find:- i) TV

ii) Abs. Error

iii) Mean Abs. Error

iv) fr. Err

v) % Err

$a_1 = 2.46$

$a_2 = 2.48$

$a_3 = 2.49$

$a_4 = 2.49$

$a_5 = 2.46$

$\Delta a_1 = |a_{mean} - a_1| = |2.48 - 2.46| = 0.02$

$\Delta a_2 = 0$

$\Delta a_3 = 0.01$

$\Delta a_4 = 0.01$

$\Delta a_5 = 0.02$

i) $a_{mean} = \frac{12.38}{5}$

$= 2.476$

$\boxed{a_{mean} \simeq 2.48}$

$\Delta a_m = \frac{\Delta a_1 + \Delta a_2 \ldots + \Delta a_5}{5}$

$\Delta a_m = \frac{0.06}{5}$

$= 0.012$

$\boxed{\Delta a_m \simeq 0.01}$

frac Error $= \frac{\Delta a_m}{a_m} = \frac{0.01}{2.48} = \frac{1}{248}$

% Error $= \frac{1}{248} \times 100 = 0.403\%$

Final measurment is reported as :-

$$a = (a_{mean} \pm \Delta a_{mean})$$

$$\boxed{a = (2.48 \pm 0.01)\ m}$$

$2.48 - 0.01 = 2.47$

$2.48 + 0.01 = 2.49$

Range or limit

i) If zero mark of a scale has been torned-off.

ii) In VERNIER CALLIPERS

if the Zeroes of MS & VS are not coinciding then zero Error may occurs.

10 VSD = 9 MSD

iii) This is Just a calibration/fabrication/manfacturing Error

★★★★★

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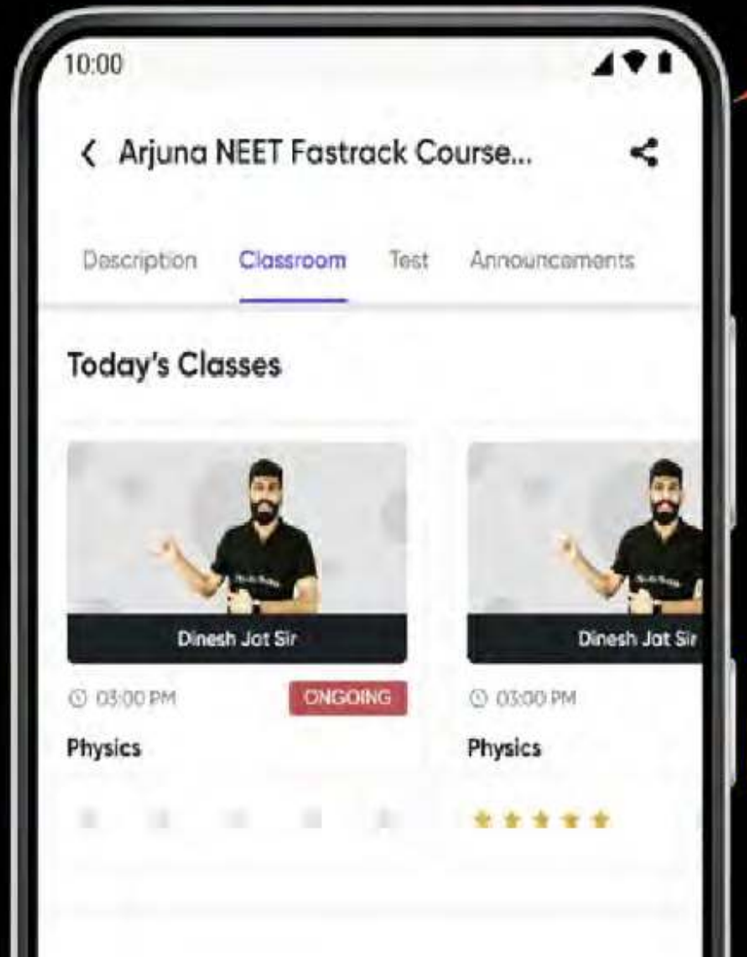
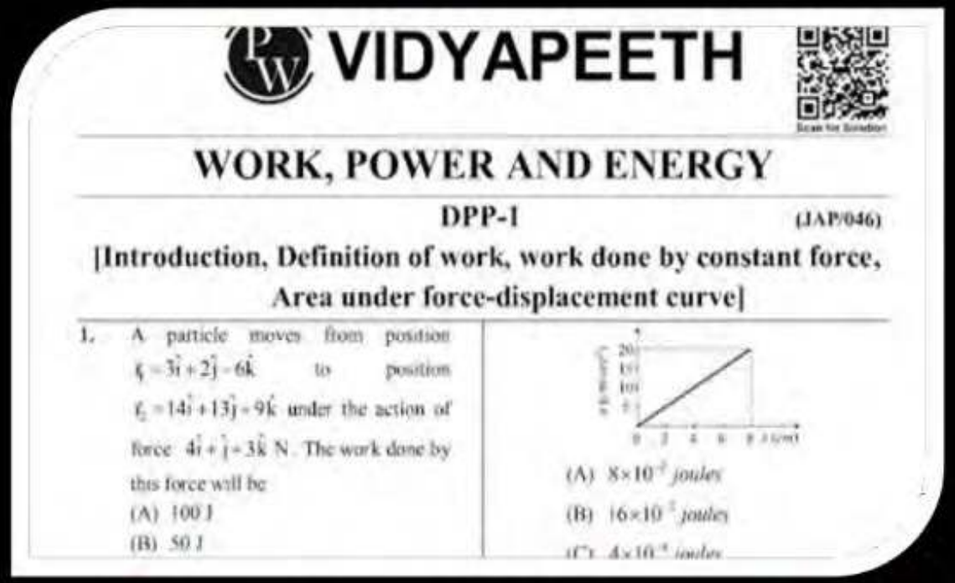


"SCAN" to join our "TELEGRAM" channel

Solve the DPP and check Solution



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WORK, POWER AND ENERGY

DPP-1 (JAP/046)

[Introduction, Definition of work, work done by constant force, Area under force-displacement curve]

1. A particle moves from position $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ to position $\vec{r}_2 = 14\hat{i} + 13\hat{j} - 9\hat{k}$ under the action of force $4\hat{i} + \hat{j} - 3\hat{k}$ N. The work done by this force will be

(A) 100 J

(B) 50 J

(A) 8×10^{-2} joules

(B) 16×10^{-2} joules

Thank You!!!!

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